

Literature Status: Endpoint Lorentz Estimates for All Cocancelling Constraints

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Status: literature already answered. The broad cocancelling-operator question in arXiv:2008.05639 is answered affirmatively. Stolyarov’s arXiv:2010.05297 proves a stronger Besov–Lorentz theorem for every closed translation- and dilation-invariant constrained space excluding point masses. For $\mathcal{W} = \ker L(D)$, exclusion of nonzero $a \otimes \delta_0$ is exactly cocancellation. Breit–Cianchi–Spector, arXiv:2512.06352, state the result directly for arbitrary homogeneous cocancelling operators and attribute the first- and higher-order cases to the intervening literature.

1 Original question

F. Hernandez and D. Spector, *Fractional Integration and Optimal Estimates for Elliptic Systems*, arXiv:2008.05639, later published in *Calculus of Variations and Partial Differential Equations* **63** (2024), Paper No. 117 [1], prove the endpoint estimate for divergence-free fields. In Section 2, arXiv PDF page 9, they isolate the broader question:

Source question. *Does the optimal endpoint Lorentz estimate hold for every cocancelling operator? Equivalently, if $L(D)$ is a homogeneous cocancelling differential operator, $0 < \alpha < d$, and $L(D)F = 0$, does one have*

$$\|I_\alpha F\|_{L^{d/(d-\alpha),1}(\mathbb{R}^d)} \leq C\|F\|_{L^1(\mathbb{R}^d)}?$$

The source says explicitly that this was not known and that an affirmative answer would resolve the open problems listed immediately beforehand.

2 Answer and exact identification

D. Stolyarov, *Hardy–Littlewood–Sobolev inequality for $p = 1$* , arXiv:2010.05297v3, later published in *Sbornik: Mathematics* **213** (2022), 844–889 [2], proves the stronger Besov–Lorentz statement. His Theorem 1.2 (arXiv PDF page 7) says that if $\mathcal{W} \subset \mathcal{S}'(\mathbb{R}^d; \mathbb{R}^\ell)$ is closed and translation- and dilation-invariant, then, for $0 < \alpha < d$,

$$I_\alpha : \mathcal{W} \cap L^1 \longrightarrow \dot{B}_{d/(d-\alpha),1}^{0,1}$$

is bounded if and only if $\mathcal{W}$ contains no nonzero $a \otimes \delta_0$. Equation (1.18) and Corollary 1.7 (arXiv PDF page 3) give the Lorentz consequence

$$\|I_\alpha F\|_{L^{d/(d-\alpha),1}} \lesssim \|F\|_{L^1}.$$

Take $\mathcal{W} = \{F : L(D)F = 0\}$. This space is closed and invariant under translations and dilations because $L(D)$ is homogeneous with constant coefficients. Moreover,

$$L(D)(a\delta_0) = 0 \iff L(\xi)a = 0 \text{ for every } \xi \neq 0 \iff a \in \bigcap_{\xi \neq 0} \ker L(\xi).$$

Thus $\mathcal{W}$ contains no nonzero point mass exactly when L is cocancelling. Stolyarov’s theorem therefore answers the source question for operators of arbitrary homogeneous order, with a stronger target norm.

This identification is also explicit in later literature. On arXiv PDF page 3 of Breit–Cianchi–Spector, *Riesz potential estimates under co-canceling constraints*, arXiv:2512.06352 [3], equation (1.8) states the desired $L^1 \rightarrow L^{n/(n-\alpha),1}$ estimate for every homogeneous cocancelling operator of order $k \in \mathbb{N}$. The following sentence attributes the first-order case to Hernandez–Raiță–Spector and the higher-order case to Stolyarov. Their Theorem 3.1 (arXiv PDF page 10) further extends the statement to general rearrangement-invariant domain and target norms.

3 Scope and search status

The all-orders Euclidean cocancelling question on arXiv PDF page 9 is completely answered affirmatively. This status does *not* by itself settle the distinct periodic-torus endpoint question mentioned on source PDF page 4.

A bounded search through 11 August 2026 checked the run registry, the local arXiv corpus, the exact question wording, the phrases “cocancelling operator” and “endpoint Lorentz”, and later papers citing the source. The decisive sources are arXiv:2010.05297 and arXiv:2512.06352. No new mathematical result is claimed in this packet.

References

- [1] F. Hernandez and D. Spector, *Fractional Integration and Optimal Estimates for Elliptic Systems*, Calc. Var. Partial Differential Equations **63** (2024), Paper No. 117; arXiv:2008.05639.
- [2] D. Stolyarov, *Hardy–Littlewood–Sobolev inequality for $p = 1$* , Sb. Math. **213** (2022), 844–889; arXiv:2010.05297.
- [3] D. Breit, A. Cianchi, and D. Spector, *Riesz potential estimates under co-canceling constraints*, arXiv:2512.06352 (2025).